



JS CheatSheet

Loops

For Loop

```
for (var i = 0; i < 10; i++) {  
    document.write(i + ": " + i*3 + "<br />");  
}  
var sum = 0;  
for (var i = 0; i < a.length; i++) {  
    sum += a[i];  
} // parsing an array  
html = "";  
for (var i of custOrder) {  
    html += "<li>" + i + "</li>";  
}
```

While Loop

```
var i = 1; // initialize  
while (i < 100) { // enters the cycle  
    i *= 2; // increment to avoid  
    document.write(i + ", "); // output  
}
```

Do While Loop

```
var i = 1; // initialize  
do { // enters cycle at  
    i *= 2; // increment to avoid  
    document.write(i + ", "); // output  
} while (i < 100) // repeats cycle if
```

Break

```
for (var i = 0; i < 10; i++) {  
    if (i == 5) { break; } // stops and exits  
    document.write(i + ", "); // last output  
}
```

Continue

```
for (var i = 0; i < 10; i++) {  
    if (i == 5) { continue; } // skips the rest  
    document.write(i + ", "); // skips 5  
}
```

Variables

```
var a; // variable  
var b = "init"; // string  
var c = "Hi" + " " + "Joe"; // "Hi Joe"  
var d = 1 + 2 + "3"; // "33"  
var e = [2,3,5,8]; // array  
var f = false; // boolean  
var g = /()/; // RegExp  
var h = function(){}; // function object  
const PI = 3.14; // constant  
var a = 1, b = 2, c = a + b; // one line  
let z = 'zzz'; // block scope local
```

Strict mode

```
"use strict"; // Use strict mode to write secure code  
= 1; // Throws an error because variable
```

Basics

On page script

```
<script type="text/javascript"> ...  
</script>
```

Include external JS file

```
<script src="filename.js"></script>
```

Delay - 1 second timeout

```
setTimeout(function () {  
  
}, 1000);
```

Functions

```
function addNumbers(a, b) {  
    return a + b; ;  
}  
x = addNumbers(1, 2);
```

Edit DOM element

```
document.getElementById("elementID").innerHTML =
```

Output

```
console.log(a); // write to the browser  
document.write(a); // write to the HTML  
alert(a); // output in an alert  
confirm("Really?"); // yes/no dialog, returns  
prompt("Your age?", "0"); // input dialog. Returns
```

Comments

```
/* Multi line  
    comment */  
// One line
```

If - Else

```
if ((age >= 14) && (age < 19)) { // logical  
    status = "Eligible."; // execute  
} else { // else block  
    status = "Not eligible."; // execute  
}
```

Switch Statement

```
switch (new Date().getDay()) { // input is condition  
    case 6: // if (day ==  
        text = "Saturday";  
        break;  
    case 0: // if (day ==  
        text = "Sunday";  
        break;  
    default: // else...  
        text = "Whatever";  
}
```

Data Types

```
var age = 18; // number  
var name = "Jane"; // string
```

Values

```
false, true // boolean
18, 3.14, 0b10011, 0xF6, NaN // number
"flower", 'John' // string
undefined, null, Infinity // special
```

Operators

```
a = b + c - d; // addition, subtraction
a = b * (c / d); // multiplication, division
x = 100 % 48; // modulo. 100 / 48 remainder =
a++; b--; // postfix increment and decrement
```

Bitwise operators

&	AND	5 & 1 (0101 & 0001)	1 (1)
	OR	5 1 (0101 0001)	5 (101)
~	NOT	~ 5 (~0101)	10 (1010)
^	XOR	5 ^ 1 (0101 ^ 0001)	4 (100)
<<	left shift	5 << 1 (0101 << 1)	10 (1010)
>>	right shift	5 >> 1 (0101 >> 1)	2 (10)
>>>	zero fill right shift	5 >>> 1 (0101 >>> 1)	2 (10)

Arithmetic

```
a * (b + c) // grouping
person.age // member
person[age] // member
!(a == b) // logical not
a != b // not equal
typeof a // type (number, object, function)
x << 2 x >> 3 // binary shifting
a = b // assignment
a == b // equals
a != b // unequal
a === b // strict equal
a !== b // strict unequal
a < b a > b // less and greater than
a <= b a >= b // less or equal, greater or equal
a += b // a = a + b (works with - * %)
a && b // logical and
```

Numbers and Math

```
var pi = 3.141;
pi.toFixed(0); // returns 3
pi.toFixed(2); // returns 3.14 - for working
pi.toPrecision(2) // returns 3.1
pi.valueOf(); // returns number
Number(true); // converts to number
Number(new Date()) // number of milliseconds since epoch
parseInt("3 months"); // returns the first number
parseFloat("3.5 days"); // returns 3.5
Number.MAX_VALUE // largest possible JS number
Number.MIN_VALUE // smallest possible JS number
Number.NEGATIVE_INFINITY // -Infinity
Number.POSITIVE_INFINITY // Infinity
```

Math.

```
var pi = Math.PI; // 3.141592653589793
Math.round(4.4); // = 4 - rounded
Math.round(4.5); // = 5
Math.pow(2, 8); // = 256 - 2 to the power of 8
Math.sqrt(49); // = 7 - square root
Math.abs(-3.14); // = 3.14 - absolute, positive
Math.ceil(3.14); // = 4 - rounded up
Math.floor(3.99); // = 3 - rounded down
Math.sin(0); // = 0 - sine
```

```
var name = {first:"Jane", last:"Doe"}; // object
var truth = false; // boolean
var sheets = ["HTML", "CSS", "JS"]; // array
var a; typeof a; // undefined
var a = null; // value
```

Objects

```
var student = { // object name
  firstName:"Jane", // list of properties
  lastName:"Doe",
  age:18,
  height:170,
  fullName: function() { // object function
    return this.firstName + " " + this.lastName;
  }
};
student.age = 19; // setting value
student[age]++; // incrementing
name = student.fullName(); // call object function
```

Strings

```
var abc = "abcdefghijklmnopqrstuvwxyz";
var esc = '\t don\'t know'; // \n new line
var len = abc.length; // string length
abc.indexOf("lmno"); // find substring
abc.lastIndexOf("lmno"); // last occurrence
abc.slice(3, 6); // cuts out "def"
abc.replace("abc", "123"); // find and replace
abc.toUpperCase(); // convert to uppercase
abc.toLowerCase(); // convert to lowercase
abc.concat(" ", str2); // abc + " " + str2
abc.charAt(2); // character at index 2
abc[2]; // unsafe, abc[2]
abc.charCodeAt(2); // character code
abc.split(","); // splitting a string
abc.split(""); // splitting on character
128.toString(16); // number to hexadecimal
```

Events

```
<button onclick="myFunction();">
  Click here
</button>
```

Mouse

onclick, oncontextmenu, ondblclick, onmousedown, onmouseenter, onmouseleave, onmousemove, onmouseover, onmouseout, onmouseup

Keyboard

onkeydown, onkeypress, onkeyup

Frame

onabort, onbeforeunload, onerror, onhashchange, onload, onpageshow, onpagehide, onresize, onscroll, onunload

Form

onblur, onchange, onfocus, onfocusin, onfocusout, oninput, oninvalid, onreset, onsearch, onselect, onsubmit

Drag

ondrag, ondragend, ondragenter, ondragleave, ondragover, ondragstart, ondrop

Clipboard

oncopy, oncut, onpaste

```
Math.cos(Math.PI); // OTHERS: tan,atan,asin,ac
Math.min(0, 3, -2, 2); // = -2 - the lowest value
Math.max(0, 3, -2, 2); // = 3 - the highest value
Math.log(1); // = 0 natural logarithm
Math.exp(1); // = 2.7182pow(E,x)
Math.random(); // random number between 0
Math.floor(Math.random() * 5) + 1; // random integ
```

Constants like Math.PI:

E, PI, SQRT2, SQRT1_2, LN2, LN10, LOG2E, Log10E

Dates

Mon Feb 17 2020 13:42:03 GMT+0200 (Eastern European Standard Time)

```
var d = new Date();
```

1581939723047 milliseconds passed since 1970

```
Number(d)
```

```
Date("2017-06-23"); //date declara
Date("2017"); // is set to Ja
Date("2017-06-23T12:00:00-09:45"); // date - time
Date("June 23 2017"); // long date fo
Date("Jun 23 2017 07:45:00 GMT+0100 (Tokyo Time)");
```

Get Times

```
var d = new Date();
```

```
a = d.getDay(); // getting the weekday
```

```
getDate(); // day as a number (1-31)
getDay(); // weekday as a number (0-6)
getFullYear(); // four digit year (yyyy)
getHours(); // hour (0-23)
getMilliseconds(); // milliseconds (0-999)
getMinutes(); // minutes (0-59)
getMonth(); // month (0-11)
getSeconds(); // seconds (0-59)
getTime(); // milliseconds since 1970
```

Setting part of a date

```
var d = new Date();
```

```
d.setDate(d.getDate() + 7); // adds a week to a dat
```

```
setDate(); // day as a number (1-31)
setFullYear(); // year (optionally month and d
setHours(); // hour (0-23)
setMilliseconds(); // milliseconds (0-999)
setMinutes(); // minutes (0-59)
setMonth(); // month (0-11)
setSeconds(); // seconds (0-59)
setTime(); // milliseconds since 1970)
```

Global Functions ()

Media

onabort, oncanplay, oncanplaythrough, ondurationchang
onended, onerror, onloadeddata, onloadedmetadata,
onloadstart, onpause, onplay, onplaying, onprogress,
onratechange, onseeked, onseeking, onstalled,
onsuspend, ontimeupdate, onvolumechange, onwaiting

Animation

animationend, animationiteration, animationstart

Miscellaneous

transitionend, onmessage, onmousewheel, ononline,
onoffline, onpopstate, onshow, onstorage, ontoggle,
onwheel, ontouchcancel, ontouchend, ontouchmove,
ontouchstart

Arrays

```
var dogs = ["Bulldog", "Beagle", "Labrador"];
var dogs = new Array("Bulldog", "Beagle", "Labrad
```

```
alert(dogs[1]); // access value at in
dogs[0] = "Bull Terier"; // change the first i
```

```
for (var i = 0; i < dogs.length; i++) { // pa
    console.log(dogs[i]);
}
```

Methods

```
dogs.toString(); // conver
dogs.join(" * "); // join:
dogs.pop(); // remove
dogs.push("Chihuahua"); // add ne
dogs[dogs.length] = "Chihuahua"; // the sa
dogs.shift(); // remove
dogs.unshift("Chihuahua"); // add ne
delete dogs[0]; // change
dogs.splice(2, 0, "Pug", "Boxer"); // add el
var animals = dogs.concat(cats,birds); //join t
dogs.slice(1,4); // elemen
dogs.sort(); // sort s
dogs.reverse(); // sort s
x.sort(function(a, b){return a - b}); // numeri
x.sort(function(a, b){return b - a}); // numeri
highest = x[0]; // first
x.sort(function(a, b){return 0.5 - Math.random()})
```

concat, copyWithin, every, fill, filter, find, findIndex,
forEach, indexOf, isArray, join, lastIndexOf, map, pop,
push, reduce, reduceRight, reverse, shift, slice, some,
sort, splice, toString, unshift, valueOf

Regular Expressions

```
var a = str.search(/CheatSheet/i);
```

Modifiers

i	perform case-insensitive matching
g	perform a global match
m	perform multiline matching

Patterns

\	Escape character
\d	find a digit
\s	find a whitespace character
\b	find match at beginning or end of a word

Errors

```
try { // block of code to
  undefinedFunction();
}
catch(err) { // block to handle
  console.log(err.message);
}
```

Throw error

```
throw "My error message"; // throw a text
```

Input validation

```
var x = document.getElementById("mynum").value; //
try {
  if(x == "") throw "empty"; //
  if(isNaN(x)) throw "not a number";
  x = Number(x);
  if(x > 10) throw "too high";
}
catch(err){ //
  document.write("Input is " + err); //
  console.error(err); //
}
finally {
  document.write("</br />Done"); //
}
```

Error name values

RangeError	A number is "out of range"
ReferenceError	An illegal reference has occurred
SyntaxError	A syntax error has occurred
TypeError	A type error has occurred
URIError	An encodeURI() error has occurred

Useful Links

n+ contains at least one n
n* contains zero or more occurrences of n
n? contains zero or one occurrences of n
^ Start of string

JSON

```
var str = '{"names":[" + // cr
'{"first":"Hakuna","lastN":"Matata" },' +
'{"first":"Jane","lastN":"Doe" },' +
'{"first":"Air","lastN":"Jordan" }]}';
obj = JSON.parse(str); // pa
document.write(obj.names[1].first); // ac
```

Send

```
var myObj = { "name":"Jane", "age":18, "city":"Ch
var myJSON = JSON.stringify(myObj);
window.location = "demo.php?x=" + myJSON;
```

Storing and retrieving

```
myObj = { "name":"Jane", "age":18, "city":"Chicag
myJSON = JSON.stringify(myObj); /
localStorage.setItem("testJSON", myJSON);
text = localStorage.getItem("testJSON"); /
obj = JSON.parse(text);
document.write(obj.name);
```

Promises

```
function sum (a, b) {
  return Promise(function (resolve, reject) {
    setTimeout(function () {
      if (typeof a !== "number" || typeof b !==
        return reject(new TypeError("Inputs must
      }
      resolve(a + b);
    }, 1000);
  });
}
var myPromise = sum(10, 5);
myPromise.then(function (result) {
  document.write(" 10 + 5: ", result);
  return sum(null, "foo"); // Invali
}).then(function () { // Won't
}).catch(function (err) { // The ca
  console.error(err); // =>Ple
});
```

States

pending, fulfilled, rejected

Properties

Promise.length, Promise.prototype

Methods

Promise.all(iterable), Promise.race(iterable),
Promise.reject(reason), Promise.resolve(value)